Notes: Foucault and Frésard (RFS, 2012)

Cross-Listing, Investment Sensitivity to Stock Price, and the Learning Hypothesis

Contents

1 Big picture

- **Question.** Does a U.S. cross-listing change how much managers rely on stock prices when making investment decisions?
- **Core hypothesis.** The paper studies the *learning hypothesis*: a U.S. cross-listing makes the firm's stock price more informative to managers, so managers place more weight on price when choosing investment.
- **Main empirical object.** The key outcome is the **investment-to-price sensitivity**: how strongly corporate investment responds to lagged Tobin's Q.
- **Main result.** Exchange-listed foreign firms have much higher investment-to-price sensitivity than firms that never cross-list. In the baseline table, $Q \times Exchange$ is positive and similar in magnitude to the standalone Q coefficient, implying that cross-listed firms' investment is roughly twice as sensitive to price.
- **Timing result.** The effect does not appear before the cross-listing date. It rises around the cross-listing year and remains visible afterward.
- **Mechanism result.** The increase is stronger when the cross-listing is more likely to increase price informativeness for managers: high firm-specific return variation, stronger U.S. information contribution, greater U.S. trading, higher analyst coverage, and more U.S. institutional ownership.
- **Alternative explanations.** The paper argues that the result is not primarily explained by governance bonding, improved disclosure, analyst forecast accuracy, or easier access to external capital.
- **Economic takeaway.** The stock market is not only a passive valuation device. Cross-listing can change the information environment of the firm, and this change can feed back into real investment decisions.

2 Incremental contribution of the paper

- **From valuation effects of cross-listing to real effects.** The cross-listing literature had mainly studied valuation, disclosure, governance, and access-to-capital channels. This paper shifts attention to a real decision: corporate investment.
- **A managerial-learning interpretation.** Instead of treating stock prices only as investor valuations, the paper studies the idea that managers learn from prices and use that information in capital allocation.
- **A direct bridge between international finance and corporate investment.** The paper connects U.S. cross-listings, international investor participation, analyst coverage, and stock-price informativeness to the sensitivity of investment to Q.
- **Cross-sectional mechanism tests.** The paper does not stop at the average $Q \times Exchange$ coefficient. It shows that the effect is larger precisely where new information production after cross-listing should be stronger.

- **Event-time evidence.** The paper shows that the investment-to-price sensitivity is not already high before cross-listing. This helps distinguish managerial learning from fixed differences between firms that choose to cross-list and firms that do not.
- **Separation from other cross-listing channels.** The paper systematically checks governance, disclosure, analyst accuracy, and financing-access explanations and finds little support for them as the main mechanism.
- **Contribution to the investment-price literature.** The paper provides international evidence that stock prices can actively guide managers' investment decisions, complementing work such as Chen, Goldstein, and Jiang (2007).

3 Data and measurement

3.1 Sample construction

- The starting point is non-U.S. firms covered by Worldscope from 1989 to 2006.
- The authors exclude financial firms, utilities, observations with missing major accounting variables, firms with assets below $10 million, and firms with negative sales.
- Cross-listing information is assembled from the Bank of New York, J.P. Morgan, Citibank, NYSE, NASDAQ, firms' annual reports, CRSP, Factiva, and LexisNexis.
- The main treated group is firms cross-listed on U.S. exchanges through Level 2 or Level 3 ADRs, ordinary listings, or New York registered shares.
- The paper also studies OTC Level 1 ADRs and Rule 144a private placements as weaker cross-listing forms.

Interpretation: The sample is designed to compare foreign firms listed on U.S. exchanges with foreign firms that never cross-list. OTC and Rule 144a listings are useful because they provide a placebo-like contrast: they are cross-listings, but should generate less trading liquidity and less new information for managers.

3.2 Key variables

- **Investment:** baseline investment is capital expenditures divided by lagged property, plant, and equipment (PPE).
- **Price signal:** the normalized stock price is Tobin's Q, computed from market equity plus book assets minus book equity, scaled by book assets.
- **Cross-listing indicator:** $Exchange_{i,t}$ equals one if firm i is listed on a U.S. exchange in year t.
- **Controls:** cash flow over assets and log total assets are included in the baseline investment regressions.
- **Performance:** later tests use return on assets (ROA) and sales growth $\Delta Sales$.

Interpretation: The central idea is that if price becomes more informative to managers, investment should load more strongly on Q. The level of investment can go up or down depending on whether price conveys good or bad news, but the *sensitivity* to price should rise.

3.3 Proxies for price informativeness

- ψ: firm-specific return variation, used as a proxy for private information in prices.
- U.S. relative industry importance: the relative importance of the firm's industry in the U.S. market compared with the home market.
- Foreign sales: the fraction of sales realized abroad.
- BKL: the Baruch, Karolyi, and Lemmon "U.S. information factor."
- U.S. trading: the fraction of trading volume occurring on U.S. exchanges.
- Coverage: average number of analyst forecasts.
- Institutional holdings: U.S. institutional ownership relative to shares outstanding.

Interpretation: These proxies try to capture whether U.S. cross-listing creates information that is new to managers. The paper cannot observe managers' information set directly, so it uses cross-sectional variation in environments where price should be more informative.

4 Models and empirical specifications

4.1 Baseline investment-to-price sensitivity model

The core specification is:

$$\begin{aligned} I_{i,t} = \alpha + \beta_0 Q_{i,t-1} + \beta_1 Exchange_{i,t-1} + \beta_2 Q_{i,t-1} \times Exchange_{i,t-1} \quad (1) \\ + \gamma_1 CF_{i,t-1} + \gamma_2 \log(TA_{i,t-1}) + \varepsilon_{i,t}. \quad (2) \end{aligned}$$

- $I_{i,t}$ is corporate investment.
- $Q_{i,t-1}$ is the lagged stock-price signal.
- $Exchange_{i,t-1}$ captures U.S. exchange cross-listing status.
- The coefficient of interest is β_2, the incremental investment-to-price sensitivity of cross-listed firms.

The marginal effect of cross-listing on investment is:

$$\frac{\partial I_{i,t}}{\partial Exchange_{i,t-1}} = \beta_1 + \beta_2 Q_{i,t-1}. \quad (3)$$

Interpretation: The learning hypothesis does not require β_1 to be positive. It predicts $\beta_2 > 0$: after cross-listing, managers should put more weight on information in prices.

4.2 Event-time design

The event-time analysis replaces the single *Exchange* indicator with dummies for years before and after the cross-listing date:

$$I_{i,t} = \alpha + \beta_0 Q_{i,t-1} + \sum_{\tau=-10}^{10} \delta_\tau \, Q_{i,t-1} \times D^{\tau}_{i,t} + Controls + \varepsilon_{i,t}. \tag{4}$$

Interpretation: If cross-listed firms simply have inherently high investment-to-price sensitivity, the coefficients should already be positive before listing. The paper instead finds the increase after the cross-listing event.

4.3 Cross-sectional mechanism tests

For each informativeness proxy, cross-listed firms are divided into *High* and *Low* groups. The regression uses:

$$I_{i,t} = \alpha + \beta_0 Q_{i,t-1} + \beta_L Q_{i,t-1} \times Low_{i,t-1} + \beta_H Q_{i,t-1} \times High_{i,t-1} + Controls + \varepsilon_{i,t}. \tag{5}$$

Interpretation: The learning hypothesis predicts $\beta_H > \beta_L$: investment should react more to price when the cross-listing creates a stronger information channel.

4.4 Performance tests

The paper also estimates whether firms with larger post-cross-listing increases in investment-to-price sensitivity subsequently perform better:

$$Performance_{i,t+k} = \alpha_i + \alpha_t + \delta_1 Pos_{i,t} + \delta_2 Neg_{i,t} + Controls + \varepsilon_{i,t+k}. \tag{6}$$

- *Pos* identifies cross-listed firms with relatively large increases in investment-to-price sensitivity.
- *Neg* identifies cross-listed firms with relatively small increases.
- Performance is measured by ROA and sales growth, both next year and over the next three years.

Interpretation: If higher sensitivity reflects useful managerial learning, it should be associated with better subsequent operating performance, especially for the *Pos* group.

5 Identification and empirical design

5.1 What the design can identify

- The paper identifies a strong association between U.S. exchange cross-listing and higher investment-to-price sensitivity.
- The strongest evidence for the learning channel comes from timing, cross-listing type, cross-sectional informativeness proxies, and alternative-explanation tests.

- The authors are careful that the design is not a randomized experiment and cannot fully rule out all omitted variables affecting both cross-listing and investment sensitivity.

5.2 Main identification concerns

- **Selection into cross-listing.** Better firms or firms with different investment policies may choose to cross-list.
- **Size and financing constraints.** Cross-listed firms are larger and may face different financing constraints.
- **Governance bonding.** Cross-listing may improve governance and make investment more responsive to market signals.
- **Disclosure and analyst coverage.** Investors may forecast investment better after cross-listing, mechanically increasing the investment-price relation.
- **Access to capital.** Cross-listing may relax financing constraints and therefore change the investment-Q relation.

5.3 How the paper addresses these concerns

- Uses firm fixed effects, Fama-MacBeth estimates, country random effects, and matched samples by size and financial constraints.
- Uses within-cross-listing-firm comparisons before and after the listing date.
- Compares exchange listings with OTC and Rule 144a listings.
- Uses multiple proxies for stock-price informativeness.
- Tests governance, disclosure, analyst forecast accuracy, and financing-access explanations directly.
- Uses a Heckman correction to address selection into exchange cross-listing.

Interpretation: The paper's design is not a clean natural experiment, but the pattern of evidence is disciplined: the main coefficient appears after cross-listing, is strongest when price should become more informative, and is not explained well by other standard cross-listing channels.

6 Tables and figures

6.1 Table 1: Sample description and Table 2: Descriptive statistics

Read:

- Table 1 shows the composition of cross-listing types and the control sample across 39 countries.
- The main exchange-listed sample contains 633 firms and 6,345 firm-year observations.
- The paper also includes 665 OTC firms, 170 Rule 144a firms, and 20,027 control firms that never cross-list in the United States.

- Table 2 shows that exchange-listed firms are much larger than control firms and have higher average Q and sales growth.
- Average Q is 1.531 for exchange-listed firms versus 1.121 for control firms.

Economic message: These tables define the empirical comparison. Cross-listed firms are not a random subset of firms; they are bigger and have higher valuations, which motivates the matching, fixed-effect, and selection tests later in the paper.

Table 1
Sample description

	Exchange		OTC		Rule 144a		Control	
	No. Firms	Obs.	No. Firms	Obs.	No. Firms	Obs.	No. Firms	Obs.
Argentina+	7	78	1	12	5	57	57	370
Australia	20	184	10	71	3	36	931	4,830
Austria	1	7	3	11	0	0	124	1,020
Belgium	2	26	0	0	0	0	140	1,219
Brazil+	21	223	3	18	3	35	298	1,972
Canada	184	1,590	148	1,265	0	0	1,076	5,476
Chile+	12	163	6	62	1	14	123	1,096
China+	12	86	20	186	4	42	1,510	7,165
Denmark	4	53	2	30	0	0	184	1,768
Finland	3	36	4	53	2	30	162	1,447
France	29	359	25	350	2	25	1,003	7,349
Germany	23	236	30	367	3	41	851	6,948
Greece+	2	15	2	10	3	19	250	826
Hong Kong	11	90	90	989	2	17	638	4,057
Hungary+	1	10	5	48	3	27	29	185
India+	9	94	5	32	45	523	572	3,396
Ireland	7	81	7	60	0	0	77	615
Israel	54	294	4	30	0	0	106	528
Italy	8	97	4	52	6	70	322	2,441
Japan	29	408	42	535	1	8	3,857	28,117
Korea+	7	66	5	32	12	131	960	6213
Mexico+	29	316	23	257	5	66	90	584
the Netherlands	31	306	19	207	1	4	221	1862
New Zealand	4	53	8	80	2	26	263	1608
Norway	5	52	1	6	0	0	105	727
Peru+	2	25	3	27	1	14	66	424
Philippines+	3	41	6	80	5	62	122	842
Portugal	2	22	4	46	2	24	79	575
Poland+	1	7	3	26	5	44	172	725
Russia+	6	38	25	127	2	11	40	82
Singapore	6	44	17	210	0	0	568	3,450
South Africa+	11	144	26	295	4	48	376	2,232
Spain	5	60	3	33	1	10	185	1,652
Sweden	9	113	9	115	1	12	363	2,519
Switzerland	8	83	5	76	1	17	240	2,188
Taiwan+	7	75	12	83	38	398	1,380	6,927
Turkey+	1	6	1	5	5	52	190	1151
the United Kingdom	54	734	76	977	1	8	2,286	16,327
Venezuela+	3	30	8	83	1	8	11	47
All countries	**633**	**6,345**	**665**	**6,946**	**170**	**1,879**	**20,027**	**130,960**

This table describes the number of cross-listed firms ("No. Firms") and the number of firm-year observations ("Obs.") in our sample classified by country of origin and type of cross-listing. It also reports the number of firms and firm-year observations in the control sample. Exchange firms are firms that are listed on a U.S. exchange (Level 2 and 3 ADRs and ordinary listings). OTC firms are firms that are listed over-the-counter as Bulletin Board or Pink Sheet issues. Rule 144a firms are firms that are listed via Rule 144a (private placement). Control firms are firms that never had stocks cross-listed in the United States. The sample period is from 1989 to 2006. +Denotes a country designated as an emerging market by Standard and Poor's Emerging Market Database.

(a) Table 1: Sample description

Table 2
Descriptive statistics

	Exchange			
Variables	Mean	Median	SD	Firm-Year
Total Assets (TA)	10,860.489	2,008.245	25,048.135	6,345
Q	1.531	1.136	1.219	6,345
Capex/PPE(t-1)	0.287	0.197	0.354	6,345
CF/TA	0.105	0.122	0.144	6,345
Δ*Sales*	0.179	0.108	0.449	6,086
	OTC			
Variables	Mean	Median	SD	Firm-Year
Total Assets (TA)	6,430.973	1,656.225	14,020.308	6,976
Q	1.241	0.972	0.949	6,976
Capex/PPE(t-1)	0.248	0.169	0.331	6,976
CF/TA	0.111	0.116	0.115	6,976
Δ*Sales*	0.147	0.093	0.410	6,293
	Rule 144a			
Variables	Mean	Median	SD	Firm-Year
Total Assets (TA)	3,588.729	1,088.569	8,735.453	1,879
Q	1.197	0.926	0.868	1,879
Capex/PPE(t-1)	0.264	0.169	0.340	1,879
CF/TA	0.128	0.124	0.082	1,879
Δ*Sales*	0.169	0.111	0.383	1,838
	Control			
Variables	Mean	Median	SD	Firm-Year
Total Assets (TA)	951.151	176.195	4,163.270	130,960
Q	1.121	0.854	0.920	130,960
Capex/PPE(t-1)	0.278	0.158	0.412	130,960
CF/TA	0.095	0.100	0.119	130,960
Δ*Sales*	0.144	0.080	0.419	128,968

This table reports the mean, median, and standard deviation of the main variables used in the analysis. All variables are defined in the appendix. We provide these statistics separately for Exchange, OTC, Rule 144a firms, and control firms. Exchange firms are firms that are listed on a U.S. exchange (Level 2 and 3 ADRs and ordinary listings). OTC firms are firms that are listed over-the-counter as Bulletin Board or Pink Sheet issues. Rule 144a firms are firms that are listed via Rule 144a (private placement). Control firms are firms that never had stocks cross-listed in the United States. The sample period is from 1989 to 2006.

(b) Table 2: Descriptive statistics

6.2 Table 3: Main results and Table 4: Robustness

Read:

- Table 3 is the central result: $Q \times Exchange$ is positive and statistically significant across many specifications.
- In the baseline specification, the coefficient on Q is 0.066 and the coefficient on $Q \times Exchange$ is also 0.066.
- Economically, this means exchange-listed firms' investment is roughly twice as sensitive to Q as the investment of non-cross-listed firms.
- The result survives firm fixed effects, Fama-MacBeth estimation, random effects, matching on size, matching on financial constraints, and a within-cross-listing-firm comparison.
- Table 4 shows that the result is robust to alternative definitions of investment, including capex over assets, capex plus R&D, and asset growth.

Economic message: The main coefficient is not a fragile accounting artifact. It is stable across estimation methods and investment measures, which makes the learning interpretation more credible.

Table 3
U.S. cross-listing and firms' investment-to-price sensitivity: Main results

	Investment (Capex over lagged PPE)							
	Baseline (1)	Firm FE (2)	F-M (3)	Country RE (4)	Match Size (5)	Match WW (6)	Match Div (7)	Cross-Listings (8)
Exchange	−0.082**	−0.063**	−0.047**	−0.051**	−0.100**	−0.091**	−0.056**	−0.109**
	[6.75]	[3.19]	[4.69]	[5.33]	[7.84]	[6.52]	[4.28]	[5.26]
Q	0.066**	0.051**	0.057**	0.076**	0.061**	0.061**	0.076**	0.726**
	[34.04]	[19.74]	[8.28]	[69.01]	[9.70]	[8.97]	[11.53]	[5.95]
Q × Exchange	0.066**	0.057**	0.060**	0.055**	0.073**	0.063**	0.056**	0.063**
	[7.26]	[5.06]	[7.74]	[10.08]	[7.51]	[5.87]	[5.60]	[4.79]
CF/TA	0.312**	0.425**	0.425**	0.311**	0.241**	0.438**	0.278**	0.169**
	[20.77]	[22.07]	[9.87]	[40.23]	[5.34]	[8.76]	[6.74]	[3.01]
log(TA)	−0.024**	−0.076**	−0.028**	−0.028**	−0.025**	−0.031**	−0.030**	−0.031**
	[24.44]	[16.81]	[10.41]	[43.86]	[9.93]	[12.06]	[10.84]	[6.29]
Country FE	Yes	No	Yes	No	Yes	Yes	Yes	Yes
Industry FE	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	No	Yes	Yes	Yes	Yes	Yes
Firm FE	No	Yes	No	No	No	No	No	No
No. firm-years	131,463	131,479	131,479	131,479	11,404	10,938	11,320	5,645
R^2/Pseudo-R^2	0.15	0.49	0.09	0.20	0.22	0.22	0.23	0.31

This table presents the estimation of Equation (1) with various estimation techniques. The dependent variable is investment, defined as capital expenditures divided by lagged property, plant, and equipment (PPE). *Exchange* is a dummy variable that is equal to one if the firm is cross-listed on a U.S. exchange and is zero otherwise. Other explanatory variables are defined in the appendix. In Column (1), we estimate Equation (1) with pooled OLS regressions with country, year, and industry fixed effects. In Column (2), we reestimate Equation (1) with firm fixed effects and without country and industry fixed effects. In Column (3), we estimate Equation (1) using Fama and MacBeth's (1973) methodology. In Column (4), we estimate Equation (1) by including country random effects. In Columns (5) to (7), we include only cross-listed firms and their matches as defined in Section 2. In Column (8), we include only cross-listing firms (before and after they cross-list). The sample period is from 1989 to 2006. The standard errors used to compute the t-statistics (in brackets) are adjusted for heteroscedasticity and within-firm clustering. **indicates statistical significance at the 1% level.

(a) Table 3: U.S. cross-listing and investment-to-price sensitivity, main results

Table 4
U.S. cross-listing and firms' investment-to-price sensitivity: Robustness

	Investment (Various measures)					
	(1)	(2)	(3)	(4)	(5)	(6)
Exchange	−0.016**	−0.009**	−0.171**	−0.016**	−0.004	−0.042**
	[4.23]	[3.25]	[3.70]	[3.20]	[0.98]	[3.33]
Q	0.007**	0.004**	0.176**	0.015**	0.010**	0.050**
	[15.52]	[11.56]	[26.50]	[22.67]	[20.03]	[28.39]
Q × Exchange	0.011**	0.006**	0.247**	0.022**	0.011**	0.038**
	[4.05]	[3.07]	[5.62]	[5.65]	[4.13]	[3.73]
CF/ TA	0.157**	0.089**	−0.309**	0.123**	0.049**	0.859**
	[44.61]	[34.88]	[6.45]	[25.80]	[13.28]	[68.69]
log(TA)	−0.002**	−0.000*	−0.045**	−0.003**	−0.001**	−0.019**
	[9.32]	[2.00]	[20.81]	[10.37]	[3.34]	[29.20]
Country FE	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
No. firm-years	136,899	136,899	131,463	136,899	136,899	149,353
R^2	0.22	0.20	0.17	0.20	0.17	0.23

In this table, we estimate Equation (1) using various measures of investment with pooled OLS regressions. In Columns (1) and (2), investment is defined as capital expenditures divided by lagged and contemporaneous total assets, respectively. In Column (3), investment is defined as capital expenditures plus R&D expenses divided by lagged PPE. In Columns (4) and (5), investment is defined as capital expenditures plus R&D expenses divided by lagged and contemporaneous total assets, respectively. Finally, in Column (6), investment is defined as the annual change in total assets divided by lagged total assets. Across all specifications, *Exchange* is a dummy variable that is equal to one if the firm is cross-listed on a U.S. exchange and zero otherwise. All other explanatory variables are defined in the appendix. The sample period is from 1989 to 2006. All estimations include country, year, and industry fixed effects. The standard errors used to compute the t-statistics (in brackets) are adjusted for heteroscedasticity and within-firm clustering. ** and * indicate statistical significance at the 1% and 5% levels,

(b) Table 4: Robustness to alternative investment measures

6.3 Figure 1: Year-by-year estimates and Figure 2: Country-by-country estimates

Read:

- Figure 1 plots the baseline investment-to-price sensitivity and the incremental sensitivity for cross-listed firms by year.
- The higher sensitivity of cross-listed firms appears throughout the sample period rather than only in one crisis or one regulatory window.
- Figure 2 plots the same idea country by country.
- The incremental sensitivity is positive in many countries, showing that the result is not purely driven by one dominant country.

Economic message: The result is broad in time and across countries. This is important because cross-listing patterns differ greatly across home markets.

cross-listings are much debated (see Doidge, Karolyi, and Stulz 2009, 2010;

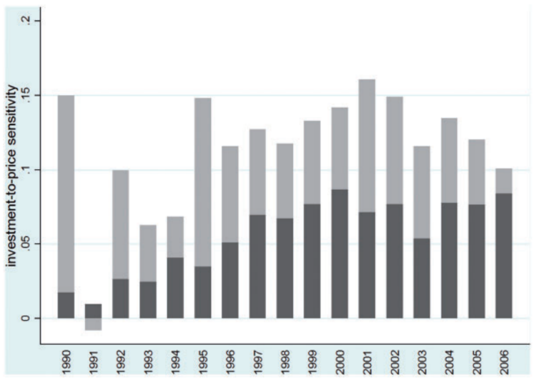

Figure 1
U.S. cross-listing and firms' investment-to-price sensitivity: Year-by-year estimations
This figure reports the results from year-by-year OLS regressions of the association between a U.S. cross-listing

(a) Figure 1: Year-by-year investment-to-price sensitivity

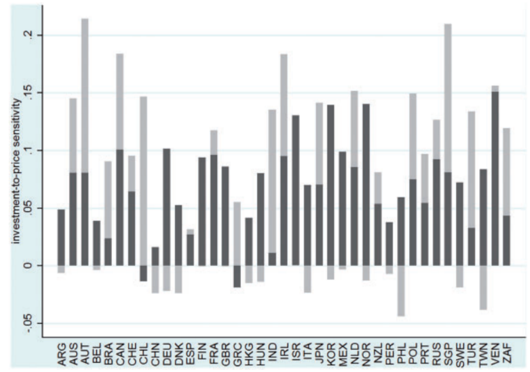

Figure 2
U.S. cross-listing and firms' investment-to-price sensitivity: Country-by-country estimation
This figure reports the results from country-by-country OLS regressions of the association between a U.

(b) Figure 2: Country-by-country investment-to-price sensitivity

6.4 Figure 3: Event-time analysis

Read:

- Figure 3 studies the evolution of $Q \times Exchange$ around the cross-listing date.
- Before cross-listing, firms that will cross-list do not have a significantly higher investment-to-price sensitivity than control firms.
- After cross-listing, the sensitivity rises and remains positive for years.
- The persistence weakens the concern that the result is driven only by a one-time event exactly at the listing date.

Economic message: The timing is central to the learning hypothesis. The effect appears when the firm's trading and information environment changes, rather than being fully pre-existing.

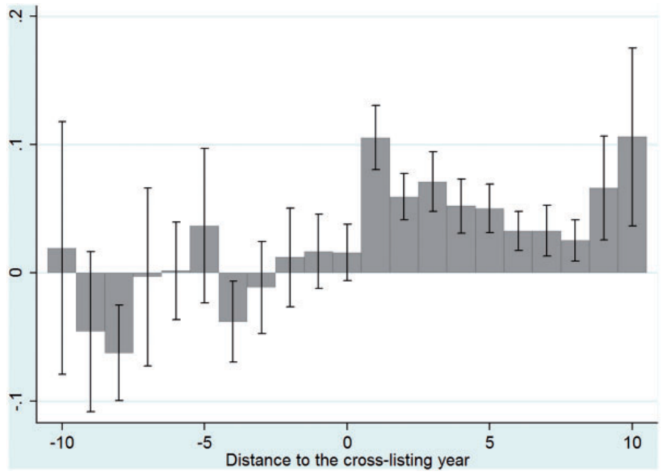

Figure 3
U.S. cross-listing and firms' investment-to-price sensitivity in event time
This figure reports results from an event-time analysis of the association between a U.S. cross-listing and firms' investment-to-price sensitivity. Specifically, we reestimate the investment model given in Equation (1) by replacing the dummy variable *Exchange* with a set of dummy variables that keep track of the number of years until or since the cross-listing date for each cross-listed firm in our sample. These dummy variables are defined over a time window of twenty years centered on the cross-listing year (year 0) for a given firm. The figure reports the coefficient estimates on the interaction between Q and each dummy variable as well as their 95% confidence interval. The sample period is from 1989 to 2006. All estimations include country, year, and industry fixed effects. The standard errors used to compute the confidence bounds are adjusted for heteroscedasticity and within-firm clustering.

Figure 3: U.S. cross-listing and investment-to-price sensitivity in event time

6.5 Table 5: Learning proxies and Table 6: Cross-listing types

Read:

- Table 5 divides cross-listed firms into high and low groups based on proxies for stock-price informativeness.
- Across the seven proxies, firms in the high-informativeness group generally show larger investment-to-price sensitivity.
- The tests use proxies such as firm-specific return variation, U.S. information contribution, foreign sales, U.S. trading, analyst coverage, and U.S. institutional holdings.
- Table 6 compares exchange listings with OTC and Rule 144a cross-listings.
- Exchange listings display the strongest investment-to-price sensitivity. OTC effects are weaker, and Rule 144a effects are mostly not different from the control group.

Economic message: These tables are the strongest mechanism evidence. If the effect were just "being cross-listed," OTC and Rule 144a listings should look similar. Instead, the effect is strongest when the cross-listing is most likely to create active trading and information production.

Table 5
Managerial learning, U.S. cross-listing, and firms' investment-to-price sensitivity sensitivity

	Investment (Capex over lagged PPE)						
	ψ (1)	U.S. Rel. Ind. (2)	Foreign Sales (3)	BKL (4)	U.S. Trading (5)	Coverage (6)	Ins. Holdings (7)
Exchange	−0.078**	−0.077**	−0.079**	−0.078**	−0.042**	−0.097**	−0.023
	[6.46]	[6.79]	[6.81]	[6.24]	[3.37]	[6.61]	[1.76]
Q	0.065**	0.066**	0.065**	0.066**	0.065**	0.065**	0.065**
	[34.02]	[34.02]	[34.07]	[33.94]	[33.52]	[33.72]	[33.13]
$Q \times Low$ (a)	0.056**	0.055**	0.057**	0.056**	0.021*	0.080**	0.004
	[5.62]	[6.02]	[6.71]	[5.44]	[2.08]	[6.09]	[0.55]
$Q \times High$ (b)	0.071**	0.077**	0.076**	0.070**	0.040**	0.035**	0.038**
	[6.87]	[7.42]	[6.46]	[6.94]	[4.62]	[3.18]	[2.85]
CF/TA	0.312**	0.312**	0.312**	0.310**	0.318**	0.316**	0.321**
	[20.74]	[20.71]	[20.76]	[20.57]	[20.87]	[20.77]	[20.72]
log(TA)	−0.024**	−0.024**	−0.024**	−0.024**	−0.024**	−0.024**	−0.024**
	[8.39]	[24.34]	[24.27]	[24.32]	[23.94]	[24.19]	[23.50]
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
No. firm-years	131,324	131,469	131,463	131,116	130,199	130,588	127,703
R^2	0.15	0.15	0.15	0.15	0.15	0.15	0.15
(a) = (b) (p-value)	0.08	0.02	0.05	0.07	0.01	0	0.01

In this table, we estimate investment Equation (1), with pooled OLS regressions, adding two interaction terms between Q and two dummy variables *Low* and *High*. The dependent variable is investment, defined as capital expenditures divided by lagged property, plant, and equipment (PPE). *Exchange* is a dummy variable that is equal to one if the firm is cross-listed on a U.S. exchange and zero otherwise. *Low* (respectively, *High*) is a dummy variable equal to one in year t for a cross-listed firm if the value of a proxy that measures the informational content of stock prices for managers is below (respectively, above) the median value of this proxy for all cross-listed firms. We use seven different firm-level variables as proxies of the informational content of stock prices for managers of cross-listed firms: (1) ψ is firm-specific return variation; (2) *U.S. Rel. Ind* is the difference in the percentage of the market capitalization of a firm's industry located in the United States and the percentage of industry market capitalization for a firm's industry in its home country; (3) *Foreign sales* measures the fraction of sales realized abroad; (4) *BKL* is the "U.S. information factor" developed by Baruch, Karolyi, and Lemmon (2007); (5) *U.S. trading* is the fraction of trading that takes place on U.S. exchanges; (6) *Coverage* is the average number of analysts issuing forecasts over a given year; and (7) *Inst. Holdings* is the fraction of U.S. institutional holdings to total shares outstanding. In the last line of the table, we report the p-value of an F-test that evaluates whether the coefficients on $Q \times Low$ and $Q \times High$ are equal. The sample period is from 1989 to 2006. All other explanatory variables are defined in the appendix. All estimations include country, year, and industry fixed effects. The standard errors used to compute the t-statistics (in brackets) are adjusted for heteroscedasticity and within-firm clustering. ** and * indicate statistical significance at the 1% and 5% levels, respectively.

(a) Table 5: Managerial learning proxies

Table 6
U.S. cross-listing and firms' investment-to-price sensitivity: Cross-listing types

	Investment (Capex over lagged PPE)							
	Baseline (1)	Firm FE (2)	F-M (3)	Country RE (4)	Match Size (5)	Match WW (6)	Match Div (7)	Cross-listings (8)
Exchange	−0.082**	−0.065**	−0.048**	−0.051**	−0.088**	−0.094**	−0.074**	−0.115**
	[6.82]	[3.29]	[4.81]	[5.43]	[7.43]	[7.45]	[6.20]	[7.13]
OTC	−0.027**	−0.044*	0.002	−0.021	−0.041**	−0.051**	−0.031**	−0.085**
	[2.60]	[2.15]	[0.03]	[1.72]	[3.59]	[4.37]	[2.77]	[4.82]
144a	0.031*	−0.032	−0.032	0.038*	0.014	0.001	0.031*	−0.025
	[2.07]	[0.70]	[0.58]	[2.45]	[0.92]	[0.11]	[2.02]	[1.23]
Q	0.066**	0.050**	0.057**	0.077**	0.065**	0.056**	0.062**	0.063**
	[34.65]	[19.59]	[8.16]	[70.51]	[13.33]	[11.09]	[13.47]	[7.35]
$Q \times Exchange$ (a)	0.065**	0.058**	0.059**	0.054**	0.064**	0.067**	0.067**	0.066**
	[7.19]	[5.25]	[7.51]	[10.03]	[6.94]	[6.76]	[7.24]	[60.9]
$Q \times OTC$ (b)	0.019*	0.038**	−0.003	0.012	0.022*	0.024**	0.024**	0.031**
	[2.19]	[3.09]	[0.34]	[1.29]	[2.34]	[2.57]	[2.59]	[2.69]
$Q \times 144a$ (c)	−0.001	0.005	0.007	−0.008	−0.006	−0.001	−0.001	−0.001
	[0.13]	[0.24]	[0.92]	[0.73]	[0.55]	[0.09]	[0.06]	[0.12]
CF/TA	0.315**	0.426**	0.429**	0.314**	0.332**	0.428**	0.340**	0.274**
	[21.34]	[22.61]	[10.04]	[41.59]	[9.21]	[11.59]	[10.97]	[6.44]
log(TA)	−0.024**	−0.076**	−0.027**	−0.027**	−0.026**	−0.026**	−0.026**	−0.033**
	[24.91]	[17.15]	[10.79]	[44.92]	[13.76]	[14.09]	[13.48]	[9.68]
Country FE	Yes	No	Yes	No	Yes	Yes	Yes	Yes
Industry FE	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	No	Yes	Yes	Yes	Yes	Yes
Firm FE	No	Yes	No	No	No	No	No	No
No. firm-years	137,142	137,158	137,158	137,158	22,768	22,202	22,767	11,384
R^2	0.15	0.49	0.09	0.21	0.19	0.19	0.19	0.24
(a) = (b) (p-value)	0.00	0.11	0.00	0.00	0.00	0.00	0.00	0.00
(a) = (c) (p-value)	0.00	0.03	0.01	0.00	0.00	0.00	0.00	0.00
(b) = (c) (p-value)	0.16	0.21	0.36	0.16	0.04	0.07	0.08	0.00

In this table, we estimate investment Equation (1) adding two interaction terms between Q and two dummy variables *OTC* and *144a* to assess whether the effect of a cross-listing on the investment-to-price sensitivity depends on the cross-listing type. The dependent variable is investment, defined as capital expenditures divided by lagged property, plant, and equipment (PPE). *Exchange* is a dummy variable that is equal to one if a firm is cross-listed on a U.S. exchange and zero otherwise. *OTC* is a dummy variable that is equal to one if a firm is cross-listed on the U.S. OTC market and zero otherwise. *144a* is a dummy variable that is equal to one if a firm is cross-listed in the United States via a Rule 144a placement. All other variables are defined in the appendix. In Column (1), we estimate Equation (1) with pooled OLS regressions with country, year, and industry fixed effects. In Column (2), we reestimate Equation (1) with firm fixed effects and without country and industry fixed effects. In Column (3), we estimate Equation (1) using Fama and MacBeth's (1973) methodology. In Column (4), we estimate Equation (1) by including country random effects. In Columns (5) to (7), we include only cross-listed firms and their matches as defined in Section 2. In Column (8), we include only cross-listing firms (before and after they cross-list) The sample period is from 1989 to 2006. We report F-tests that evaluate whether

(b) Table 6: Cross-listing types

6.6 Table 7: Ex post performance and Table 8: Self-selection

Read:

- Table 7 tests whether cross-listed firms with large increases in investment-to-price sensitivity subsequently perform better.
- The positive-sensitivity group has stronger subsequent ROA and sales growth, especially for sales growth.
- This supports the interpretation that managers are using useful information in prices, rather than merely reacting mechanically to market valuations.
- Table 8 uses a Heckman two-stage procedure to address self-selection into U.S. exchange cross-listing.
- The second-stage $Q \times Exchange$ coefficient remains positive and significant.

Economic message: Table 7 gives the real-efficiency implication: learning from prices should improve capital allocation. Table 8 shows that the main result survives an explicit correction for the cross-listing selection process.

Table 7
Investment-to-price sensitivity and ex post performance

	Panel A: Performance (Next year)				Panel B: Performance (Average of next 3 years)			
	ROA (1)		Δ*Sales* (2)		*ROA* (3)		Δ*Sales* (4)	
Exchange	0.012		0.069**		0.012		0.037*	
	[1.94]		[3.09]		[1.82]		[2.15]	
Pos (a)		0.032**		0.109**		0.017*		0.064**
		[4.45]		[4.03]		[2.32]		[3.27]
Neg (b)		0.002		0.048*		0.009		0.022
		[0.26]		[2.07]		[1.40]		[1.30]
log(TA)	−0.030**	−0.030**	−0.145**	−0.145**	−0.036**	−0.036**	−0.175**	−0.175**
	[20.94]	[20.97]	[27.21]	[27.22]	[24.03]	[24.03]	[39.62]	[39.65]
Debt / TA	−0.009	−0.009	−0.028	−0.027	0.037**	0.037**	0.011	0.011
	[1.86]	[1.77]	[1.54]	[1.49]	[7.69]	[7.72]	[0.78]	[0.83]
Cash /TA	0.072**	0.071**	0.149**	0.149**	0.034**	0.034**	0.168**	0.168**
	[9.78]	[9.74]	[5.45]	[5.43]	[4.61]	[4.60]	[8.33]	[8.31]
PPE /TA	0.005	0.005	−0.023	−0.023	0.022**	0.022**	−0.008	−0.008
	[0.74]	[0.75]	[0.86]	[0.85]	[3.45]	[3.45]	[0.38]	[0.38]
Firm/Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
No. firm-years	131,153	131,153	130,089	130,089	124,799	124,799	126,545	126,545
R^2	0.56	0.56	0.35	0.35	0.74	0.74	0.60	0.60
(a) = (b) (*p*-value)	0.00	0.00		0.00		0.05		

This table presents the results of OLS regressions of the relationship between a U.S. cross-listing and firms' ex post performance. Performance is defined as one-year-ahead (three-years-ahead) return on asset (*ROA*) or sales growth (Δ*Sales*). *Exchange* is a dummy variable that is equal to one if the firm is cross-listed on a U.S. exchange and zero otherwise. *Pos* (*Neg*) is a dummy variable that is equal to one if a cross-listed firm experiences a relatively large (small) increase in its investment-to-price sensitivity after cross-listing in the United States. The text and the online appendix detail the computation of these two dummy variables. All other variables are defined in the appendix. The sample period is from 1989 to 2006. We report the F-tests that evaluate whether

(a) Table 7: Investment-to-price sensitivity and ex post performance

Table 8
U.S. cross-listing and firms' investment-to-price sensitivity: Self-selection

	Heckman	
	(First-Stage) Probit	Second Stage
Exchange		−0.087**
		[4.25]
Q		0.064**
		[32.61]
Q × *Exchange*		0.063**
		[6.64]
CF / TA		0.348**
		[23.14]
log(TA)	0.359**	−0.023**
	[64.05]	[20.83]
Debt / TA	−0.197**	
	[3.05]	
External Dependence	0.002*	
	[2.37]	
Δ*Sales*	0.057**	
	[2.69]	
Median Industry Q	−3.894**	
	[3.68]	
ROA	−0.348**	
	[5.07]	
Foreign Sales	1.101**	
	[39.07]	
Common Law	0.790**	
	[35.96]	
Market Capitalization	−0.360**	
	[20.00]	
Inverse Mills Ratio		0.007
		[0.77]
Industry and Year FE	Yes	Yes
No. firm-years	156,982	131,221
Pseudo-R^2/R^2	0.41	0.15

In this table, we estimate the investment Equation (1) with the Heckman (1979) two-stage estimation procedure. The first column reports the results of the (first-stage) probit estimation, where the dependent variable is *Exchange*, a dummy variable that is equal to one if the firm is cross-listed on a U.S. exchange and zero otherwise. The second column reports the (second-stage) OLS estimates of Equation (1) in which the dependent variable is investment defined as capital expenditures divided by lagged property, plant, and equipment (PPE) and in which we include the *Inverse Mills Ratio* computed using the first-stage probit estimates to account for self-selection. All other explanatory variables are defined in the appendix. The sample period is from 1989 to 2006. The standard errors used to compute the t-statistics (in brackets) are adjusted for heteroscedasticity and within-firm clustering. ** and * indicate statistical significance at the 1% and 5% levels, respectively.

(b) Table 8: Self-selection

6.7 Table 9: Governance and Table 10: Disclosure / forecast accuracy

Read:

- Table 9 tests whether governance improvements explain the higher investment-to-price sensitivity.
- The evidence is not consistent with governance as the main channel: the sensitivity is often stronger for firms or countries where governance improvement should be smaller.
- Table 10 tests whether disclosure quality and analyst forecast accuracy explain the effect.
- The paper finds no systematic evidence that firms with larger disclosure or forecast-accuracy improvements have higher investment-to-price sensitivity.

Economic message: These tables help separate managerial learning from two standard cross-listing mechanisms. Cross-listing may improve governance and disclosure, but those channels do not appear to be the main reason investment becomes more sensitive to price.

Table 9
Alternative explanation: Change in governance quality

	Panel A: Country-Level Tests								Panel B: Firm-Level	
	Anti-Self-Dealing		Legal Origin		GDP per Capita		Market Capitalization		GOV	ΔGOV
	Low (a)	High (b)	Low (a)	High (b)	Low (a)	High (b)	Low (a)	High (b)		
Exchange	−0.048**	−0.099**	−0.055**	−0.098**	−0.067**	−0.082**	−0.055**	−0.089**	−0.094**	−0.080*
	[3.27]	[7.92]	[4.16]	[6.79]	[3.84]	[7.39]	[3.75]	[7.03]	[2.97]	[2.71]
Q	0.066**	0.064**	0.061**	0.066**	0.041**	0.071**	0.056**	0.070**	0.047**	0.050*
	[37.91]	[42.27]	[42.59]	[34.83]	[18.30]	[53.65]	[34.20]	[43.86]	[5.05]	[5.44]
$Q \times$ *Exchange*	0.032**	0.080**	0.040**	0.081**	0.043**	0.068**	0.036**	0.075**		
	[3.27]	[12.17]	[4.51]	[11.02]	[3.84]	[11.03]	[3.72]	[11.02]		
$Q \times$ *Low* (a)									0.060*	0.086*
									[2.40]	[2.82]
$Q \times$ *High* (b)									0.075**	0.044*
									[2.82]	[2.00]
Control variables	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country/industry/year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
No. firm-years	63,749	67,056	86,559	44,246	32,781	98,688	61,470	69,335	3,799	3,799
R^2	0.18	0.13	0.17	0.13	0.13	0.16	0.16	0.15	0.22	0.22
Q: (a) = (b)	0.43		0.14		0.00		0.00			
$Q \times$ *Exchange*: (a) = (b)	0.00		0.00		0.06		0.00			
(a) = (b)									0.37	0.04

This table presents estimates of the link between a U.S. cross-listing and firms' investment-to-price sensitivity (Equation (1)) for different levels of governance quality. The depender is investment, defined as capital expenditures divided by lagged property, plant, and equipment (PPE). *Exchange* is a dummy variable that is equal to one if the firm is cross-listed exchange and zero otherwise. In Panel A, we partition countries based on four proxies for governance quality and financial development: the anti-self-dealing and legal origin ind Djankov et al. (2008), and GDP per capital and market capitalization from the Worldbank. For each variable, we assign a country to the *Low* group if the value of the proxy for thi is below the sample median for this proxy and to the *High* group otherwise. We estimate the investment Equation (1) via a seemingly unrelated regression (SUR) system that com *Low* and *High*. The SUR estimation provides the joint-variance-covariance matrix that we use to test the cross-equation restrictions that appear at the bottom of the table (we p-value of these tests). The sample period is from 1989 to 2006. In Panel B, we partition cross-listed firms using firm-level measures from the RiskMetrics database. *GOV* is a g index built using forty-one governance attributes from the RiskMetrics database (see the text). Δ*GOV* represents the average change of *GOV* over the period 2003–2006. *Low* (res *High*) is a dummy variable equal to one in year t for a cross-listed firm if the value of *GOV* or Δ*GOV* is below (respectively, above) the median value of this proxy for all cross-lis In the last line of the table, we report the p-value of an F-test that evaluates whether the coefficients on $Q \times$ *Low* and $Q \times$ *High* are equal. The sample period is from 2003 to

(a) Table 9: Governance quality alternative explanation

Table 10
Alternative explanation: Change in investor's ability to predict future investment

	Panel A: Country-Level Tests				Panel B: Firm-Level Tests		
	Disclosure (DLSW)		Disclosure (JM)		S&P Score	Accuracy	ΔAccuracy
	Low (a)	High (b)	Low (a)	High (b)			
Exchange	−0.050**	−0.095**	−0.059**	−0.082**	−0.117**	−0.069**	−0.094**
	[5.07]	[7.35]	[3.80]	[6.19]	[3.46]	[4.85]	[3.45]
Q	0.066**	0.069**	0.048**	0.071**	0.058**	0.065**	0.064**
	[27.51]	[45.21]	[31.77]	[42.72]	[19.52]	[33.71]	[33.20]
$Q \times$ *Exchange*	0.039**	0.076**	0.045**	0.069**			
	[4.72]	[11.19]	[4.19]	[9.74]			
$Q \times$ *Low* (a)					0.072**	0.067**	0.050*
					[4.44]	[5.64]	[2.19]
$Q \times$ *High* (b)					0.060*	0.045**	0.082**
					[2.02]	[4.07]	[2.79]
Control variables	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country/industry/ year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
No. firm-years	61,167	63,401	57,657	68,028	19,162	130,323	127,274
R^2	0.16	0.15	0.14	0.14	0.21	0.15	0.14
Q: (a) = (b)	0.24		0.01				
$Q \times$ *Exchange*: (a) = (b)	0.00		0.03				
(a) = (b)					0.57	0.11	0.16

This table presents estimates of the link between a U.S. cross-listing and firms' investment-to-price sensitivity (Equation (1)) for different measures of information quality. The dependent variable is investment, defined a capital expenditures divided by lagged property, plant, and equipment (PPE). *Exchange* is a dummy variable tha is equal to one if the firm is cross-listed on a U.S. exchange and zero otherwise. In Panel A, we partition countrie based on two indices of disclosure quality: the disclosure index from Djankov et al. (2008) (Disclosure DLSV and the disclosure index from Jin and Myers (2006) (Disclosure (JM)). For each index, we assign a country to the *Low* group if it has a value below the sample median for this variable and to the *High* group otherwise We estimate the investment Equation (1) via a seemingly unrelated regression (SUR) system that combine the *Low* and *High*. The SUR estimation provides the joint-variance-covariance matrix that we use to test the cross-equation restrictions that appear in the last two lines of the table (we report the p-value of these tests) The sample period is from 1989 to 2006. In Panel B, we partition cross-listed firms based on three firm-leve proxy for their informational environment. The *S&P Score* measures corporate disclosure practice to measure firm-level disclosure quality from Standard & Poor's (available only in 2000 or 2001). *Accuracy* measures the precision of analyst forecasts. Δ*Accuracy* measures the change in forecast accuracy around the cross-listing event (three years before and after the cross-listing). In the last line of the table, we report the p-value of ar F-test that evaluates whether the coefficients on $Q \times$ *Low* and $Q \times$ *High* are equal. All explanatory variables are defined in the appendix. All estimations include country, year, and industry fixed effects as well as *CF/TA* anc *log(TA)* as control variables. The standard errors used to compute the t-statistics (in brackets) are adjusted fo heteroscedasticity and within-firm clustering. ** and * indicate statistical significance at the 1% and 5% levels respectively

(b) Table 10: Disclosure and forecast-accuracy alternative explanation

6.8 Table 11: Access to external finance

Read:

- Table 11 tests whether improved access to external capital explains the higher investment-to-price sensitivity.
- The paper compares Level 2 versus Level 3 ADRs, capital raising around cross-listing, changes in size, changes in the Whited-Wu index, and changes in dividends.
- The $Q \times High$ and $Q \times Low$ coefficients are not statistically different in these tests.
- This weakens the interpretation that the higher investment-to-price sensitivity is simply a financing-constraint story.

Economic message: The access-to-capital channel is plausible because cross-listing can make equity issuance easier. But the evidence here suggests that easier financing is not the main driver of the investment-to-price sensitivity result.

Table 11
Alternative explanation: Change in firms' access to external finance

	Investment (Capex over lagged PPE)				
	Level 2 vs. 3 (1)	Cap. Raising (2)	ΔSize (3)	ΔWW (4)	ΔDIV (5)
Exchange	-0.079^{**}	-0.082^{**}	-0.102^{**}	-0.102^{**}	-0.096^{**}
	[6.79]	[6.64]	[7.40]	[7.42]	[7.12]
Q	0.065^{**}	0.065^{**}	0.065^{**}	0.065^{**}	0.065^{**}
	[34.07]	[34.06]	[33.71]	[33.71]	[33.69]
$Q \times Low$ (a)	0.064^{**}	0.059^{**}	0.074^{**}	0.061^{**}	0.058^{**}
	[7.11]	[6.47]	[6.56]	[3.36]	[5.23]
$Q \times High$ (b)	0.074^{**}	0.074^{**}	0.063^{**}	0.074^{**}	0.073^{**}
	[4.45]	[5.97]	[4.49]	[6.53]	[6.25]
Control variables	Yes	Yes	Yes	Yes	Yes
Country/industry/ year FE	Yes	Yes	Yes	Yes	Yes
No. firm-years	131,363	131,363	129,271	129,271	129,271
R^2	0.15	0.15	0.15	0.15	0.14
F-test: (a)-(b) (p-value)	0.49	0.16	0.38	0.36	0.18

In this table, we estimate the investment Equation (1) with pooled OLS regressions adding interaction term between Q and two dummy variables *High* and *Low* that measure the change in the intensity of financial constraint for cross-listed firms. The dependent variable is investment, defined as capital expenditures divided by lagge property, plant, and equipment (PPE). *Exchange* is a dummy variable that is equal to one if the firm is cross-liste on a U.S. exchange and zero otherwise. In Column (1), *High* is a dummy variable equal to one for Level 3 AD firms. In Column (2), firms that do augment their capital issuance around the cross-listing are in the *High* group In Column (3), firms whose size increases are in the *Low* group. In Column (4), firms whose Whited and W index decreases are in the Low group. In Column (5), firms whose dividend payout increases are in the *Hig* group. In the last line of the table, we report the p-value of an F-test that evaluates whether the coefficients o $Q \times High$ and $Q \times Low$ are equal. All variables are defined in the appendix. All estimations include country year, and industry fixed effects. The standard errors used to compute the t-statistics (in brackets) are adjusted fo

Table 11: Change in firms' access to external finance

6.9 Appendix Table A1: Variable definitions and sources

Read:

- Appendix Table A1 defines the main firm-level and country-level variables.
- It is especially useful for interpreting the informativeness proxies, governance measures, disclosure measures, and financing-constraint proxies.

Economic message: The appendix table clarifies that the paper combines accounting data, market data, institutional ownership, analyst coverage, governance indices, disclosure measures, and cross-listing records.

Appendix: Definitions and sources of the variables

Table A1

This table provides definitions and sources of all the variables used in the analysis.

Variables	Definition	Source
Firm-level variables		
Exchange	Dummy variable that takes one if a firm is cross-listed on a U.S. exchange (level 2 and 3 ADRs and ordinary listings) and is zero otherwise	Various sources (See Section 1)
OTC	Dummy variable that takes one if a firm is cross-listed over-the-counter (level 1 ADR) and is zero otherwise	Various sources (See Section 1)
144*a*	Dummy variable that takes one if a firm is cross-listed via a Rule 144a (private placement) and is zero otherwise	Various sources (See Section 1)
Capex	Capital expenditures (in million USD)	Worldscope
(*Tobin's*)*Q*	(Book value of assets – book value of equity + market value of equity) / book value of assets	Worldscope
PPE	Property, plant, and equipment	Worldscope
Total assets (*TA*)	Book value of total assets	Worldscope
CF/TA	Cash flows from operations over total assets	Worldscope
Δ*Sales*	Percentage change in (inflation-adjusted) sales over year $t-2$ to t	Worldscope
R&D	R&D expenses. Set to zero if missing	Worldscope
Debt	Total debt (long term plus short term)	Worldscope
Cash	Sum of cash and short term investments	Worldscope
ROA	Sum of earnings before interest, taxes, depreciation, and amortization over total assets	Worldscope
Foreign Sales	Proportion of sales generated from operations in foreign countries over total sales	Worldscope
Ins. Holdings	Proportion of shares held by U.S. institutions as a fraction of common shares outstanding	CDA/Spectrum (SEC 13(f) filings)
U.S. Trading	Proportion of the total volume that takes place on U.S. markets defined as the trading volume ($) on U.S. exchange divided by the total (domestic and U.S.) volume ($)	Datastream and CRSP
BKL	"U.S. information factor" developed by Baruch, Karolyi, and Lemmon (2007)	Datastream and CRSP
Coverage	Number of analysts issuing at least one earnings forecasts over the year	I/B/E/S International summary files
Ψ	Firm specific return variation computed as $\psi_{i,t} = ln[(1 - R^2_{i,t})/R^2_{i,t}]$, where $R^2_{i,t}$ represents the R^2 from a regression of firm$i$ weekly returns on both the local and U.S. market returns in year t. The local and U.S. market indices are value-weighted and exclude the firm in question	Datastream
GOV	Governance index based on forty-one attributes on board, audit, antitakeover, compensation, and ownership developed by Aggarwal et al. (2011)	RiskMetrics (Sample restricted to the 2003–2006 period)

(a) Appendix Table A1, part 1

Table A1
Continued

Variables	Definition	Source
Δ*GOV*	Average annual change in *GOV* for cross-listed firms over the 2003–2006 period	
S&P Score	Index that counts whether a firm discloses information that would be relevant to investors (ranges from zero to ninety-one)	Standard & Poor's (Sample restricted to the 2000–2001 period)
Accuracy	Annual average of the negative of the absolute forecast error (absolute value of the difference between the estimated earnings and the actual earnings, divided by the actual earnings)	I/B/E/S International detail files
Δ*Accuracy*_	Difference between the average *Accuracy* over the three years that follow the cross-listing and the average *Accuracy* over the three years that precede it	I/B/E/S International detail files
Capital Raising	Difference between to total capital raised over the three years that follow the cross-listing (including all public and private equity and debt issued at home, in the United States, as well as in other markets) and the total capital raised over the three years that precede it	Securities Data Corporation (SDC)
WW	Index measuring the severity of financial constraints developed by Whited and Wu (2006)	Worldscope
DIV	Dummy variable that equals one if a firm pays dividend and is zero otherwise	Worldscope

(b) Appendix Table A1, part 2

7 Short summary

- The paper studies whether cross-listing in the United States changes how managers use stock prices in corporate investment decisions.
- The sample includes non-U.S. Worldscope firms from 1989 to 2006, including 633 exchange-listed firms and a large control sample of never-cross-listed firms.
- The baseline result is that exchange-listed firms have about twice the investment-to-price sensitivity of non-cross-listed firms.
- The result is robust to firm fixed effects, Fama-MacBeth estimates, matching, alternative investment definitions, and selection corrections.
- Event-time evidence shows that the effect appears after cross-listing, not before.
- Cross-sectional evidence shows that the effect is stronger when cross-listing is more likely to increase price informativeness for managers.
- OTC and Rule 144a cross-listings have weaker effects, consistent with the idea that liquidity and information production matter.
- Firms with larger increases in investment-to-price sensitivity show better subsequent operating performance.

- Governance, disclosure, analyst accuracy, and access-to-capital alternatives do not explain the main pattern well.

8 Conclusion

8.1 Core conclusion

Foucault and Frésard (2012) argue that U.S. cross-listing changes the real role of stock prices inside the firm. After cross-listing, managers appear to rely more on stock prices when making investment decisions because prices contain more information that is useful to them.

8.2 Economic intuition

Investors may observe signals about a firm's growth opportunities, product-market prospects, foreign demand, or strategic value. When a firm cross-lists in the United States, more informed investors can trade the stock, more analysts may cover the firm, and U.S. market information may become embedded in the price. Managers can then use this price signal as an additional source of information when allocating capital.

8.3 Incremental contribution revisited

The paper contributes by showing that cross-listing has a real corporate-investment implication, not only a valuation or governance implication. It also contributes to the managerial-learning literature by using cross-listings as a setting where the informativeness of prices plausibly changes.

8.4 Implications

- For corporate finance, the paper suggests that financial markets can influence real investment through information feedback, not only through financing costs.
- For international finance, the paper shows that cross-listing can change internal firm decisions by changing who trades and what information is reflected in prices.
- For empirical work, the paper highlights that investment-Q sensitivity can contain information about managerial learning, not only financing constraints or market mispricing.

8.5 Limitations

- The paper cannot directly observe the information that managers learn from stock prices.
- The cross-listing decision is endogenous, so the evidence is not a randomized causal design.
- The paper relies on proxies for price informativeness, governance quality, disclosure, analyst accuracy, and financing constraints.
- Some omitted factors may still jointly affect the decision to cross-list and the investment-to-price sensitivity.

8.6 Takeaway

The paper's main message is that stock prices can become more useful to managers after a U.S. cross-listing. Cross-listing expands the information environment, and managers appear to use that information when making real investment decisions.